

CELIA RUBIO-MADRIGAL

I am a PhD candidate at CISPA, where I study challenges in graph learning regarding the dual role of graphs as data and computation structures. I hold degrees in Mathematics and Computer Science from UCM, and a "la Caixa" Foundation Fellowship that funded my master's at Strathclyde. I am also IT for the European Girls' Mathematical Olympiad.

EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security 2023-now
 - Relational Machine Learning Lab – Supervisor: Dr. Rebekka Burkholz
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid 2021-22
 - "Identification of Boolean functions that need big circuits" – Supervisor: Dr. Ismael Rodríguez Laguna
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) 2021
 - "Differential geometry methods in ML applications" – Supervisor: Dr. David Martín de Diego
- **Consultant Developer intern** – Guidewire Software – Node.js search engine for internal raised issues 2021

EDUCATION



- **PhD in Computer Science** – CISPA & Universität des Saarlandes (Saarbrücken, Germany) 2023-now
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) 2022-23
 - Fully funded by "la Caixa" Fellowship – Earned the Department's Best Overall Prize.
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 2017-22
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 2017-22

PUBLICATIONS



- A.G. Reddy, R. Verma, C.R.M., R. Burkholz, K. Muandet. *Predictive Sufficiency for OOD Generalization via Gradient Boosting*. 14th International Conference on Learning Representations (ICLR), 2026.
- A.G. Reddy, C.R.M., R. Burkholz, K. Muandet. *When Shift Happens - Confounding Is to Blame*. 14th International Conference on Learning Representations (ICLR), 2026.
- T. Jacobs, A. Gadhikar, C.R.M., R. Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity*. 14th International Conference on Learning Representations (ICLR), 2026.
- C.R.M., R. Burkholz. *Fixed Aggregation Features Can Rival GNNs*. 20th Women in Machine Learning Workshop (WiML) at NeurIPS, 2025.
- C.R.M.*, A. Jamadandi*, R. Burkholz. *GNNs Getting ComFy: Community and Feature Similarity Guided Rewiring*. 13th International Conference on Learning Representations (ICLR), 2025.
- A. Jamadandi*, C.R.M.*, R. Burkholz. *Spectral Graph Pruning Against Over-Squashing and Over-Smoothing*. 38th Conference on Neural Information Processing Systems (NeurIPS), 2024.

TALKS



- **Invited talks** – Seminars and reading groups 2025
 - GLOW reading group, TUWien ML Research Unit seminar (in person), Cohere For AI reading group.
- **Contributed talks** – Conferences and workshops 2024-25
 - Lightning talks: GTML 2025, MLG 2025 (two), NetSci 2025. Contributed talk: HAICON 2024.

AWARDS



• Travel

- Women in Machine Learning Travel Awards: *To attend the 38th and 39th NeurIPS in Canada & US* 2024-25
- SAP Grant: *To attend the 9th Heidelberg Laureate Forum in Germany* 2022

• Postgraduate

- Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* 2023
- 🏆 Nova III Student List: *Spain's top 10 talents under 25 in Computer Science* 2023
- 9th Heidelberg Laureate Forum, with interviews: 📺 HLFF Spotlight, 📺 Tribuna Complutense 2022
- 🏆 "la Caixa" Foundation Fellowship, with interviews: 📺 El País, 📺 El Plural, 📺 TeleMadrid 2022

• Undergraduate

- 🏆 Aula de Verano "Blas Cabrera" for Spain's top undergraduate records: *Intl. Univ. Menéndez Pelayo (UIMP)* 2022
- 🏆 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
- Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22

• Pre-undergraduate

- 🏆 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017
- 🏆 Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* 2017
- 📺 Leading Program "la Caixa": *Madrid's top 20 university admission scores* 2017
- 10/10 with Honours at IES San Mateo: *Top 5 students at Spain's only High School of Excellence* 2017
- ESTALMAT: *2-year mathematics talent program for top 25 high school students in Madrid* 2012-14

TEACHING



- **Master's and Bachelor's thesis advising** – CIPA & Universität des Saarlandes 2024-now
- **Math Olympiad Trainer** – International EGMO Preparatory Seminar in Bonn, Germany 2026
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – University of Strathclyde 2023
- "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel 📧

SERVICE



- **Academic service** 2024-now
 - Reviewer for MLG @ ECML PKDD 2025, ICLR 2026, CPAL 2026, ICML 2026.
 - Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Math Olympiad service**
 - 🏆 IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers 📧 2023-now
 - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
 - Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. 2020-24
- **Student representation** – Facultad de Informática (Fdi), Universidad Complutense de Madrid 2020-22
 - Secretary of the Fdi Student Council 2021-22. Member of the Fdi Faculty Board 2020-22 (during COVID).
- **L^AT_EX package development: ddphonism** 2019
 - 🏆 Generates twelve-tone music diagrams with TikZ. Included in MiKTeX and TeXLive; try it out in Overleaf.

LANGUAGES



Spanish – Native

English – TOEFL iBT 110/120 (2021)

German – Goethe-Zertifikat B1 (2017)